

mental to many aspects of reasoning capabilities. Methods for identifying mechanisms that LMs implement, especially when these differ from human-designed algorithms, can provide crucial insights on how to build more robust LMs, control their behavior, and predict their failures. Our experiments focus on small-scale models, and we do not anticipate any immediate ethical considerations associated with our findings.

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References

- Barrington, D. A. Bounded-width polynomial-size branching programs recognize exactly those languages in NC1. *Journal of Computer and System Sciences*, 38(1):150–164, 1989. ISSN 0022-0000. doi: [https://doi.org/10.1016/0022-0000\(89\)90037-8](https://doi.org/10.1016/0022-0000(89)90037-8). URL <https://www.sciencedirect.com/science/article/pii/0022000089900378>.
- Bender, E. M. and Koller, A. Climbing towards NLU: On meaning, form, and understanding in the age of data. In Jurafsky, D., Chai, J., Schluter, N., and Tetreault, J. (eds.), *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 5185–5198, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.463. URL <https://aclanthology.org/2020.acl-main.463/>.
- Bhattacharya, S., Ahuja, K., and Goyal, N. On the Ability and Limitations of Transformers to Recognize Formal Languages. In Webber, B., Cohn, T., He, Y., and Liu, Y. (eds.), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7096–7116, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.576. URL <https://aclanthology.org/2020.emnlp-main.576/>.
- Biderman, S., Schoelkopf, H., Anthony, Q. G., Bradley, H., O’Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pp. 2397–2430. PMLR, 2023.
- Chen, A., Shwartz-Ziv, R., Cho, K., Leavitt, M. L., and Saphra, N. Sudden Drops in the Loss: Syntax Acquisition, Phase Transitions, and Simplicity Bias in MLMs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=M05PiKHELW>.
- Clark, K., Khandelwal, U., Levy, O., and Manning, C. D. What does BERT look at? an analysis of BERT’s attention. In Linzen, T., Chrupala, G., Belinkov, Y., and Hupkes, D. (eds.), *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 276–286, Florence, Italy, August 2019. Association for Computational Linguistics. doi: 10.18653/v1/W19-4828. URL <https://aclanthology.org/W19-4828/>.
- Delétang, G., Ruoss, A., Grau-Moya, J., Genewein, T., Wengliang, L. K., Catt, E., Cundy, C., Hutter, M., Legg, S., Veness, J., and Ortega, P. A. Neural Networks and the Chomsky Hierarchy. In *11th International Conference on Learning Representations*, 2023.
- Heimersheim, S. and Nanda, N. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024.
- Hu, M. Y., Chen, A., Saphra, N., and Cho, K. Latent state models of training dynamics. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL <https://openreview.net/forum?id=NE2xXWo0LF>.
- Huet, A., Houdi, Z. B., and Rossi, D. Episodic memories generation and evaluation benchmark for large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=6ycX677p2l>.
- jin04, JackS, Karvonen, A., and Can. OthelloGPT learned a bag of heuristics, 2024. URL <https://www.lesswrong.com/posts/gcpNuEZnxAPayaKBY/othellogpt-learned-a-bag-of-heuristics-1>.
- Karvonen, A. Emergent World Models and Latent Variable Estimation in Chess-Playing Language Models. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=PPTrmvEnPW>.
- Kim, N. and Schuster, S. Entity Tracking in Language Models. In Rogers, A., Boyd-Graber, J., and Okazaki, N. (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3835–3855, Toronto, Canada, July 2023. Association for Computational Linguistics.

- doi: 10.18653/v1/2023.acl-long.213. URL <https://aclanthology.org/2023.acl-long.213/>.
- Li, B. Z., Nye, M., and Andreas, J. Implicit Representations of Meaning in Neural Language Models. In Zong, C., Xia, F., Li, W., and Navigli, R. (eds.), *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pp. 1813–1827, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.143. URL <https://aclanthology.org/2021.acl-long.143/>.
- Li, K., Hopkins, A. K., Bau, D., Viégas, F., Pfister, H., and Wattenberg, M. Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=DeG07_TcZvT.
- Liu, B., Ash, J. T., Goel, S., Krishnamurthy, A., and Zhang, C. Transformers Learn Shortcuts to Automata. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=De4FYqjFueZ>.
- McCoy, R. T., Min, J., and Linzen, T. BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance. In Alishahi, A., Belinkov, Y., Chrupała, G., Hupkes, D., Pinter, Y., and Sajjad, H. (eds.), *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 217–227, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.blackboxnlp-1.21. URL <https://aclanthology.org/2020.blackboxnlp-1.21/>.
- Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in GPT. *Advances in Neural Information Processing Systems*, 35:17359–17372, 2022.
- Merrill, W. and Sabharwal, A. The Parallelism Tradeoff: Limitations of Log-Precision Transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023. doi: 10.1162/tacl.a.00562. URL <https://aclanthology.org/2023.tacl-1.31/>.
- Merrill, W. and Sabharwal, A. A Little Depth Goes a Long Way: The Expressive Power of Log-Depth Transformers. In *NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning*, 2024.
- Merrill, W., Sabharwal, A., and Smith, N. A. Saturated Transformers are Constant-Depth Threshold Circuits. *Transactions of the Association for Computational Linguistics*, 10:843–856, 2022. doi: 10.1162/tacl.a.00493. URL <https://aclanthology.org/2022.tacl-1.49/>.
- Merrill, W., Petty, J., and Sabharwal, A. The Illusion of State in State-Space Models. In *Forty-first International Conference on Machine Learning*, 2024. URL <https://openreview.net/forum?id=QZgo9JZpLq>.
- Merullo, J., Eickhoff, C., and Pavlick, E. Circuit Component Reuse Across Tasks in Transformer Language Models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=fpoAYV6Wsk>.
- Nanda, N., Lee, A., and Wattenberg, M. Emergent Linear Representations in World Models of Self-Supervised Sequence Models, 2023. URL <https://arxiv.org/abs/2309.00941>.
- Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Johnston, S., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. In-context Learning and Induction Heads. *Transformer Circuits Thread*, 2022. <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html>.
- Prakash, N., Shaham, T. R., Haklay, T., Belinkov, Y., and Bau, D. Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking. In *Proceedings of the 2024 International Conference on Learning Representations*, 2024. arXiv:2402.14811.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., Sutskever, I., et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- Shi, X., Padhi, I., and Knight, K. Does string-based neural MT learn source syntax? In Su, J., Duh, K., and Carreras, X. (eds.), *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 1526–1534, Austin, Texas, November 2016. Association for Computational Linguistics. doi: 10.18653/v1/D16-1159. URL <https://aclanthology.org/D16-1159/>.
- Strobl, L., Merrill, W., Weiss, G., Chiang, D., and Angluin, D. What Formal Languages Can Transformers Express? A Survey. *Transactions of the Association for Computational Linguistics*, 12:543–561, 2024. doi: 10.1162/tacl.a.00663. URL <https://aclanthology.org/2024.tacl-1.30/>.

- Vafa, K., Chen, J. Y., Rambachan, A., Kleinberg, J., and Mullainathan, S. Evaluating the World Model Implicit in a Generative Model. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=aVK4JFpegY>.
- Wang, K. R., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=NpsVSN6o4u1>.
- Yang, S., Gribovskaya, E., Kassner, N., Geva, M., and Riedel, S. Do Large Language Models Latently Perform Multi-Hop Reasoning? In Ku, L.-W., Martins, A., and Srikumar, V. (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 10210–10229, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.550. URL <https://aclanthology.org/2024.acl-long.550/>.
- Zhang, F. and Nanda, N. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Hf17y6u9BC>.
- Zhong, Z., Liu, Z., Tegmark, M., and Andreas, J. The clock and the pizza: Two stories in mechanistic explanation of neural networks. *Advances in Neural Information Processing Systems*, 36, 2024.